Scenario -> File Upload

❌ Weak: "Test the search feature."

✅ Search feature test successful!

I ran a web search for OpenAI's official website and received results.

- Search: OpenAI official website
- Status: Successful
- Results found: Multiple relevant pages
- Source verification: Results include OpenAI's official website and Help Center.

Using ICEPOT method: ->

I:-> Write Test Cases to test the Search Feature in the e-commerce application

C: -> Search Different Category Products like meat, dairy, etc. Search with name, Search with SUPC, Sort the search result

E: -> Consider example like -> Search a vegetable, Search a fruit wit its SUPC

P: -> Act as a Test Engineer

O: -> Provide output in the form of a table with proper headers like, Test Case ID, Pre-condition, Test Case name, Description, Steps, Expected Result, etc.

T: -> Test Cases should have the maximum coverage

O/P using above Prompt ->

As a Test Engineer, I have designed a comprehensive test suite to thoroughly validate the search functionality of your e-commerce application. This includes searching across various categories (such as meat, dairy, vegetables, and fruits), searching by

product name and SUPC (Store/Shop Unique Product Code), sorting options, and handling edge cases.

Here is the structured test case matrix for maximum coverage:

E-Commerce Search Feature Test Cases

Test Case ID	Test Case Name	Pre-condition	Description	Steps	Expected Result
TC_SRC_001	Search Dairy Product by Name	User is on the e-commerce home page.	Verify that searching for a dairy product by name displays relevant results.	1. Click on the search bar. 2. Type "Milk" and press Enter.	Search results page should display products strictly matching or related to "Milk" within the Dairy category.
TC_SRC_002	Search Meat Product by Name	User is on the e-commerce home page.	Verify that searching for a meat product by name returns	1. Click on the search bar. 2. Type "Chicken Breast" and click Search.	Search results page should list fresh and frozen chicken breast

Test Case ID	Test Case Name	Pre-condition	Description	Steps	Expected Result
			accurate results.		products with correct pricing and availability.
TC_SRC_003	Search Vegetable by Name	User is on the e-commerce home page.	Verify that searching for a vegetable by name returns accurate catalog items.	1. Click on the search bar. 2. Type "Broccoli" and press Enter.	Search results should display fresh broccoli items along with relevant imagery and pricing.
TC_SRC_004	Search Fruit by SUPC	User is on the e-commerce home page, valid fruit SUPC	Verify that searching for a specific fruit using its unique SUPC code	1. Click on the search bar. 2. Enter a valid fruit SUPC (e.g., SUPC12345 for Organic Apple).	The application should directly navigate to the product detail page or display

Test Case ID	Test Case Name	Pre-condition	Description	Steps	Expected Result
		is known.	returns a single exact match.	3. Submit the search.	the exact fruit matching the SUPC.
TC_SRC_005	Search Vegetable by SUPC	User is on the e-commerce home page, valid vegetable SUPC is known.	Verify that searching for a specific vegetable using its unique SUPC code returns an exact match.	1. Click on the search bar. 2. Enter a valid vegetable SUPC (e.g., SUPC98765 for Carrot bundle). 3. Submit the search.	The application should display the exact vegetable item associated with that SUPC.
TC_SRC_006	Sort Search Results: Price Low to High	Search results are displayed for a broad category (e.g., "Dairy").	Verify that sorting search results by price in ascending order	1. Perform a search for "Dairy". 2. Locate and click the Sort dropdown.	Products should be dynamically re-ordered so that the cheapest items

Test Case ID	Test Case Name	Pre-condition	Description	Steps	Expected Result
			functions correctly.	3. Select "Price: Low to High".	appear first.
TC_SRC_007	Sort Search Results: Price High to Low	Search results are displayed for a broad category (e.g., "Meat").	Verify that sorting search results by price in descending order functions correctly.	1. Perform a search for "Meat". 2. Select Sort -> "Price: High to Low".	Products should be re-ordered with the highest-priced items appearing first.
TC_SRC_008	Sort Search Results: Relevance / Popularity	Search results are displayed.	Verify that default or popularity-based sorting highlights trending or most relevant items.	1. Perform a search for "Vegetables". 2. Select Sort -> "Popularity" or "Relevance".	Results should display items ranked by customer purchase frequency or search match relevance.

Test Case ID	Test Case Name	Pre-condition	Description	Steps	Expected Result
TC_SRC_009	Search with Non-Existent Name (Negative)	User is on the home page.	Verify the system behavior when a searched product name does not exist.	1. Type a random string (e.g., "Xyzabc123") in the search bar. 2. Press Enter.	A friendly "No results found" message should appear, optionally with recommended trending products or categories.
TC_SRC_010	Search with Invalid SUPC (Negative)	User is on the home page.	Verify system response when an invalid or expired SUPC is entered.	1. Type an invalid SUPC (e.g., SUPC00000) in the search bar. 2. Press Enter.	The system should display a clear error or "Product not found" message without breaking the layout.

Test Case ID	Test Case Name	Pre-condition	Description	Steps	Expected Result
TC_SRC_011	Autocomplete and Suggestions Dropdown	User is on the home page.	Verify that typing a product prefix provides real-time autocomplete suggestions.	1. Click on the search bar. 2. Type "Che" (for Cheese/Cherries).	A dropdown list with predictive search terms, matching categories, and product thumbnails should appear instantly.
TC_SRC_012	Special Characters and SQL Injection Test (Security)	User is on the home page.	Verify that the search input safely sanitizes special characters and script tags.	1. Enter special strings like <script>alert(1)</script> or '; DROP TABLE products;-- into the search bar. 2. Execute search.	The system should sanitize the input, prevent code execution, and safely return either no results or sanitized text.

Additional Test Engineering Recommendations:

- **Performance Testing: Ensure search query response time remains under 500ms even with high database concurrent loads.**

- **Fuzzy Search Validation: Test spelling mistakes (e.g., searching "Tomatoe" instead of "Tomato") to check if typo-tolerance algorithms correctly fetch intended items.**